



Title

lpcde — Local polynomial conditional density estimation and inference.

Syntax

```
lpcde yvar xvars [if] [in] [, grid(var) xeval(numlist) bw(var or #) p(#) q(#)
prbc(#) qrbc(#) mu(#) nu(#) norbc kernel(KernelFn) bwselect(BwMethod)
covflag(CovMethod) ng(#) gridspacing(Spacing) nonneg normalize genvars(prefix)
rgrid(var) rindex(var) level(#) separator(#) plot graph_options(GraphOpts) ]
```

Description

lpcde implements local polynomial conditional CDF, density, and derivative estimation. The first variable in *yvar* *xvars* is the outcome; remaining variables are the conditioning covariates. Numerical calculations are implemented in standalone Mata code and are intended to match the companion R and Python packages.

Companion command: **lpbwcode** for bandwidth selection.

Options

Estimation

grid(*var*) specifies the y-grid on which the conditional density is estimated. When omitted, the R package default grid is used.

xeval(*numlist*) specifies the conditioning value. The list must contain one number for each conditioning covariate. When omitted, componentwise medians are used.

bw(*var*|#) specifies a bandwidth variable or a positive scalar bandwidth. When omitted, **lpbwcode** is called internally with **bwselect**() .

p(#) and **q**(#) specify local polynomial orders in the y- and x-directions. Defaults are **p**(2) and **q**(1) .

prbc(#) and **qrbc**(#) specify robust bias-correction polynomial orders. Defaults are **p**()+1 and **q**()+1 .

mu(#) and **nu**(#) specify derivative orders with respect to y and x. Defaults are **mu**(1) and **nu**(0) subject to the polynomial orders.

norbc suppresses robust bias correction and centers confidence intervals at the conventional estimate.

kernel(*KernelFn*) specifies **epanechnikov**, **triangular**, or **uniform**. Default is **epanechnikov** .

covflag(*CovMethod*) specifies covariance computation: **full**, **diag**, or **off**. Default is **full** .

nonneg truncates conventional point estimates below at zero. **normalize** normalizes conventional estimates to integrate to one over the y-grid.

Bandwidth Selection

bwselect(*BwMethod*) specifies **imse-rot** or **mse-rot**. Default is **imse-rot**; ignored when **bw**() is supplied.

ng(#) and **gridspacing**(*Spacing*) control automatic grid generation when **grid**() is omitted. **gridspacing**(**quantile**) requests quantile spacing.

Storing and Displaying Results

genvars(*prefix*) stores double-precision results in variables named *prefix_grid*, *prefix_bw*, *prefix_nh*, *prefix_est*, *prefix_est_rbc*, *prefix_se*, *prefix_se_rbc*, *prefix_ci_l*, and *prefix_ci_r*.

rgrid(*var*) displays rows nearest to the supplied y-grid. **rindex**(*var*) displays rows nearest to supplied row indices.

level(#) controls pointwise confidence interval level. Default is **level(95)**.

separator(#) draws a separator line after every # rows in the display.

Plotting

plot plots the conventional point estimate with pointwise confidence intervals. **graph_options()** passes options to **twoway**.

Saved Results

lpcde saves the following in **e()**:

e(result)	matrix with columns y_grid bw nh est est_RBC se se_RBC
	CI_l CI_r index
e(CovMat)	conventional covariance matrix
e(CovMat_RBC)	robust bias-corrected covariance matrix
e(N)	sample size
e(d)	number of conditioning covariates
e(p), e(q), e(mu), e(nu)	polynomial and derivative orders

Examples

```
. set obs 500
. set seed 42
. gen double x = rnormal()
. gen double y = x + rnormal()
. gen double ygrid = -2 + (_n-1) if _n <= 5
. lpcde y x, grid(ygrid) xeval(0) bw(1) genvars(cde)
```

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